

IBM Cognos for Student Success

Speaking script for the 6-slide presentation. Team: Ajesh, Dhanush, Balaji and Jeeva. About 4 minutes at a calm pace.

Slide 1: Cover

Good morning everyone. We are Ajesh, Dhanush, Balaji and Jeeva. Our topic is data warehousing with IBM Cognos.

IBM's motto is one word: THINK. So we have built this talk the way IBM itself works, in three steps: Observe, Reflect and Make. First we observe a real problem. Then we reflect on a solution. Then we make it, using IBM's own tools.

Slide 2: The problem (Observe)

Let us observe the problem. A university cannot tell which students are falling behind until the final results come out.

There are three reasons. First, the data is scattered: attendance, internal marks, fees and library records sit in separate systems. Second, reports are slow: staff combine spreadsheets by hand, so the report comes only at the end of the semester. Third, there is no early warning: a student who needs help is noticed only after failing an exam.

So the university has all the data, but not in one place.

Slide 3: The solution (Reflect)

Now the solution. We use three IBM products as one pipeline.

Step one, Collect: IBM DataStage copies data from every campus system each night. This is called ETL, which means extract, transform and load.

Step two, Store: IBM Db2 Warehouse keeps all that data clean and in one place. This is our data warehouse.

Step three, Analyse: IBM Cognos Analytics reads the warehouse and builds reports and dashboards.

Step four, Decide: teachers and department heads open one dashboard and take action.

Slide 4: The warehouse design (Reflect)

This is how the warehouse is designed. In the middle is the fact table, called Student progress. It holds the numbers: attendance, marks and fees due.

Around it are four dimension tables: Time, Subject, Student and Teacher. They describe each row.

Because the shape looks like a star, it is called a star schema. It is simple to understand, and Cognos works fast on it.

Slide 5: The output in IBM Cognos (Make)

Now we make it. In Cognos we connect to the warehouse, give the tables simple business names, and drag and drop the fields onto a dashboard. No SQL coding is needed.

This is the kind of dashboard Cognos gives us. These are sample figures for illustration, not real university data.

At the top, three tiles show the main numbers: students at risk, average attendance and average internal mark.

The bar chart shows which department has the most students at risk. The line chart shows the attendance trend over six months.

A teacher can click on any bar to see the list of students.

Slide 6: The result

What changes for the university? Before, results were known only after the exam, each office saw only its own records, and struggling students got help too late.

After, the dashboard is ready every morning, every student record is linked, and teachers can reach a struggling student in time. These are the expected benefits of this design.

We end where IBM begins, with one word: THINK. The warehouse and Cognos give teachers trusted data, so they can think better and help every student succeed.

Thank you. We are happy to take questions.

If the judges ask questions

Question	Short answer
What is a data warehouse?	A central store of historical data from many systems, organised for analysis and reporting.
What is ETL?	Extract, transform, load: taking data out of source systems, cleaning it, and loading it into the warehouse.
What is a star schema?	A design with one fact table of numbers in the middle and dimension tables around it that describe those numbers.
What is IBM Cognos?	IBM's business intelligence tool that turns warehouse data into reports and dashboards.
How is a student marked "at risk"?	In this example, a student whose attendance is below 75%. The rule can be changed in the Cognos model.
Is the dashboard real?	No, it is a mock-up with sample figures to show what Cognos would produce.